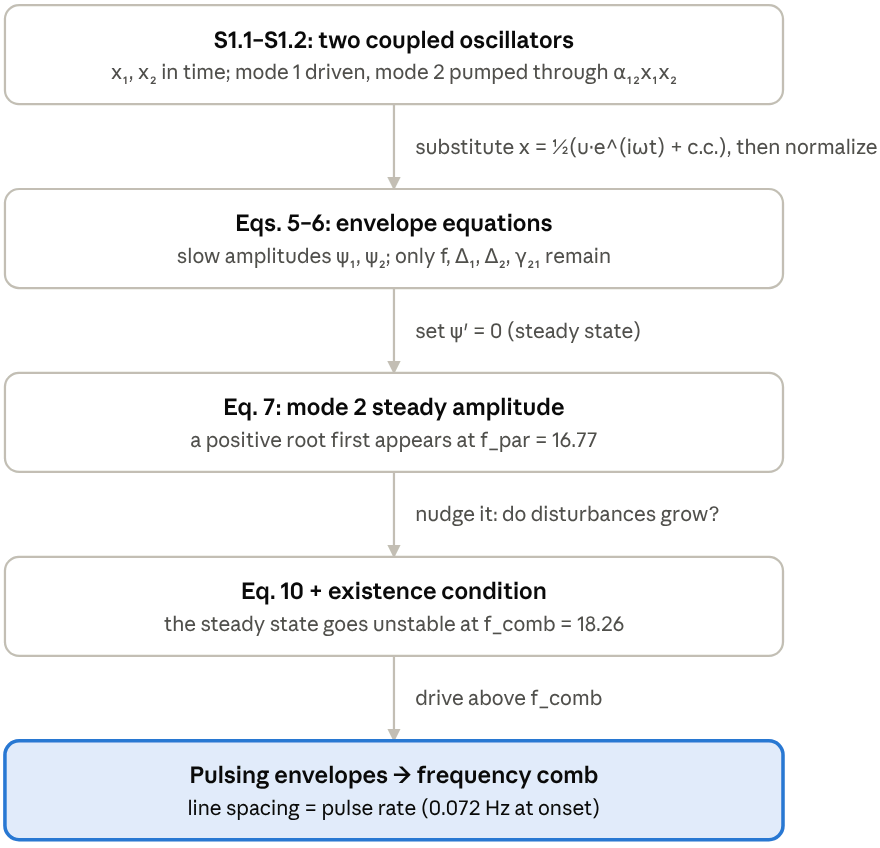
Phononic Frequency Combs — Equation Walkthrough

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Roadmap

Each equation comes from the one before it; the arrows say what operation takes you to the next step.

Each equation is one step from the physics to the comb threshold



equation chain · S1.1–S1.2 to Eq. 10

The thresholds use the paper's baseline values ($\Delta_1 = 5$, $\Delta_2 = -6.5$, $\gamma_{21} = 1$), as in `coupled_modes.ipynb`.

Parameter glossary

The key distinction: **f is how hard you push; Δ_1 is where (in frequency) you push.** Baseline values are the ones used in `coupled_modes.ipynb`.

Symbol	Name	What it physically is	Units	Baseline
x_1, x_2	displacement	how far each mode is deflected at time t	length	—
ω_1, ω_2	natural frequencies	each mode's own resonant frequency	rad/s	62.8, 32.4
γ_1, γ_2	damping rates	how fast each mode loses energy	1/s	0.105, 0.105
F	drive amplitude	how hard you push mode 1 (physical)	force/mass	—
ω_D	drive frequency	the frequency you push at	rad/s	63.35
α_{12}, α_{22}	coupling	how strongly the modes affect each other	—	100 (equal)
u_1, u_2	complex amplitudes	slowly varying amplitude and phase of each mode	length	—
ψ_1, ψ_2	normalized amplitudes	u rescaled so 1 means coupling ≈ damping	none	—
f	normalized drive	drive strength, not a frequency	none	10–20
Δ_1	mode 1 detuning	$(\omega_D - \omega_1)/\gamma_1$: drive offset from mode 1, in units of γ_1	none	5
Δ_2	mode 2 detuning	$(\omega_D - 2\omega_2)/(2\gamma_1)$: half-drive offset from mode 2, in units of γ_1	none	−6.5
κ	mistuning	$\Delta_2 - \Delta_1/2$: how far ω_2 is from $\omega_1/2$	none	−9
γ_{21}	damping ratio	γ_2/γ_1	none	1

Δ is a detuning (a frequency difference), not a delta function. One unit of Δ is γ_1, the half-width of mode 1's peak; the full width at half maximum is $2\gamma_1$.

Step 1: The physics (S1.1–S1.2)

Two coupled oscillators; only mode 1 is driven, and mode 2 gets all its energy from mode 1 through the coupling.

```
\ddot{x}_1 + 2\gamma_1\dot{x}_1 + \omega_1^2 x_1 + \alpha_{22}x_2^2 =
F\cos(\omega_D t)
```

```
\ddot{x}_2 + 2\gamma_2\dot{x}_2 + \omega_2^2 x_2 + \alpha_{12}x_1x_2 = 0
```

Term	What it does	If you increase it
$\ddot{x}$	inertia	—
$2\gamma\dot{x}$	damping: removes energy every cycle	the mode dies out faster; everything needs more drive
$\omega^2 x$	spring: sets the natural frequency	the mode rings at a higher frequency
F cos(ω_D t)	the drive, applied only to mode 1	mode 1 gets bigger
$\alpha_{12}x_1x_2$	mode 1 pumps mode 2	mode 2 turns on at lower drive
$\alpha_{22}x_2^2$	mode 2 pushes back on mode 1	mode 1 is drained harder

Parametric pumping. Rewrite mode 2's equation to see that mode 1 modulates mode 2's stiffness:

```
\ddot{x}_2 + 2\gamma_2\dot{x}_2 + \left[\omega_2^2 +
\alpha_{12}x_1(t)\right]x_2 = 0
```

Since x_1 oscillates at ω_D ≈ $2\omega_2$, the stiffness is pumped at twice mode 2's frequency, like pumping a swing by standing and squatting twice per swing. The energy gained each cycle is proportional to x_2, so it multiplies what is already there: mode 2 needs a small starting value (x2_0 = 1e-4 in the notebook) and grows exponentially once pumping beats damping.

Pump depletion. Unlike a motor-driven swing demo, the pump here is itself a resonator with limited energy, and the $\alpha_{22}x_2^2$ term lets mode 2 drain it. That two-way exchange is what makes the comb possible.

Step 2: The substitution (S1.3–S1.4)

Each mode is split into a fast carrier (a known function) times a slow complex amplitude u (the unknown).

```
x_1 = \tfrac{1}{2}\left(u_1 e^{i\omega_D t} + u_1^* e^{-i\omega_D t}\right)
```

```
x_2 = \tfrac{1}{2}\left(u_2 e^{i\omega_D t/2} + u_2^* e^{-i\omega_D t/2}\right)
```

- **Mode 1's carrier is ω_D** because it is forced at the drive frequency.
- **Mode 2's carrier is ω_D/2** because pumping at ω_D makes mode 2 respond at half that. This is why the FFT shows mode 2 locked at f_D/2 = 5.042 Hz, not its natural 5.15 Hz.
- Writing $u_1 = |u_1|e^{i\varphi_1}$ gives $x_1 = |u_1| \cos(\omega_D t + \varphi_1)$: $|u_1|$ is the envelope in the plots, φ_1 the phase.

In signal-processing terms this is complex baseband (I/Q demodulation): u is the I/Q signal and e^{iω_D t} the local oscillator. Tracking u instead of x lets the solver step on the $1/\gamma_1$ timescale instead of every carrier cycle, which was about 29× faster for Case C at Q_1 = 300.

Step 3: Envelope equations (paper Eqs. 5–6)

Substituting Step 2 into Step 1, keeping only terms at each mode's carrier and dropping ü (u changes slowly), then normalizing, gives the paper's Eqs. (5)–(6). An independent derivation from S1.1–S1.2 gives the same equations term for term.

```
\frac{\partial \psi_1}{\partial \tau} = -if - (1 + i\Delta_1)\psi_1 + i\psi_2^2
```

```
\frac{\partial \psi_2}{\partial \tau} = -(\gamma_{21} + i\Delta_2)\psi_2 + 2i\psi_1\psi_2^*
```

Term	Meaning
−i·f	the drive feeding mode 1

Term	Meaning
$-1\cdot\psi_1$ and $-\gamma_{21}\cdot\psi_2$	damping: shrinks the magnitude (mode 1 at rate 1, mode 2 at γ_{21}, in units of γ_1)
$-i\Delta_1\psi_1$ and $-i\Delta_2\psi_2$	detuning: rotates the phase without changing the magnitude
$+i\psi_2^2$	back-action: mode 2 drains mode 1 (pump depletion)
$+2i\psi_1\psi_2^*$	pumping: mode 1 feeds mode 2

Damping vs. detuning. $-(1 + i\Delta_1)\psi_1$ distributes to $-1\cdot\psi_1 - i\Delta_1\cdot\psi_1$; the 1 and Δ_1 are added, not multiplied. Alone, $\psi' = -\psi$ gives $\psi_0 e^{-\tau}$ (shrinks) and $\psi' = -i\Delta\psi$ gives $\psi_0 e^{-i\Delta\tau}$ (rotates, same size). Together they trace an inward spiral on the complex plane. They come from different terms in S1.1: damping from $2\gamma_1\dot{x}_1$, detuning from the mismatch $\omega_1^2 - \omega_D^2 \approx -2\omega_D\gamma_1\Delta_1$.

Why the complex conjugate matters. The ψ_2 conjugate in the pumping term makes it phase-sensitive: one phase of mode 2 grows, the opposite phase shrinks, as on a swing pumped at the wrong moment. That is the signature of parametric gain.

Step 4: Why normalize

After normalizing, f_1, Q_1 and α drop out completely; only four numbers set the physics.

```
\tau = \gamma_1 t,\quad f = \frac{\alpha_{12}}{8\gamma_1^2\omega_D^2}F,\quad
\psi_1 = \frac{\alpha_{12}}{4\gamma_1\omega_D}u_1,\quad \psi_2 =
\frac{\sqrt{\alpha_{12}\alpha_{22}}}{4\gamma_1\omega_D}u_2
```

Knob	Plain meaning
f	the amplitude mode 1 would reach if driven exactly on resonance with no coupling: $\psi_1 = f/(i - \Delta_1)$, so the magnitude of ψ_1 equals f at $\Delta_1 = 0$
Δ_1	how many γ_1 the drive sits from mode 1
$\Delta_2 = \Delta_1/2 + \kappa$	how many γ_1 half the drive sits from mode 2
γ_{21}	how lossy mode 2 is compared with mode 1

- **$\psi \approx 1$** means the coupling is about as strong as the damping; below that the modes barely interact.
- **Scaling knobs (f_1, Q_1, α)** change only units: Hz, seconds and force. The thresholds stay at f_par = 16.77 and f_comb = 18.26 whatever they are set to.
- **Required force scales as $\gamma_1^2\omega_D^2/\alpha$.** At fixed Q that is $\propto f_1^4$: doubling f_1 from 10 to 20 Hz takes F for f = 20 from 70 to 1127.
- **Comb spacing scales with $\gamma_1 = \omega_1/(2Q_1)$**: higher Q gives more closely spaced lines (0.070 Hz at $Q_1 = 300$, 0.021 Hz at $Q_1 = 1000$).
- The notebook sets $\alpha_{12} = \alpha_{22} = \alpha$, so $\sqrt{(\alpha_{12}\alpha_{22})} = \alpha$ and both modes share one scale.

Step 5: Steady states and Eq. 7

Setting $\psi_1' = \psi_2' = 0$ gives two kinds of steady state; mode 2 switches on at f_par = 16.77 for the baseline.

Solution A, mode 2 off ($\psi_2 = 0$). An ordinary driven oscillator; the $(1 + \Delta_1^2)$ factor is the cost of driving off resonance. At f = 10 this gives $10/\sqrt{26} = 1.96$; the simulation gave 1.97.

```
\psi_1 = \frac{f}{i - \Delta_1} \quad\Rightarrow\quad |\psi_1| = \frac{f}
{\sqrt{1 + \Delta_1^2}}
```

Solution B, mode 2 on ($\psi_2 \neq 0$). Mode 2's equation fixes mode 1's amplitude at exactly the level where pumping balances mode 2's damping plus mistuning. This is saturation: driving harder leaves mode 1 at 3.29 and sends the extra energy to mode 2. The simulation gave 3.34.

```
|\psi_{1P}| = \tfrac{1}{2}\sqrt{\gamma_{21}^2 + \Delta_2^2}
```

Eq. 7. Substituting Solution B back into mode 1's equation gives mode 2's steady amplitude:

```
|\psi_{2P}|^4 + (\gamma_{21} - \Delta_1\Delta_2)|\psi_{2P}|^2 + \tfrac{1}{4}
(1 + \Delta_1^2)(\gamma_{21}^2 + \Delta_2^2) - f^2 = 0
```

Only even powers of the magnitude appear, so set $x = |\psi_2 P|^2$ (a substitution, not an approximation; x must be real and ⩾ 0). It becomes a quadratic:

```
x^2 + b\,x + (c_0 - f^2) = 0,\qquad b = \gamma_{21} - \Delta_1\Delta_2,\qquad
c_0 = \tfrac{1}{4}(1+\Delta_1^2)(\gamma_{21}^2+\Delta_2^2)
```

- Baseline: $b = 1 - (5)(-6.5) = 33.5$ and $c_0 = ¼(26)(43.25) = 281.125$.
- The roots sum to $-b < 0$ and multiply to $c_0 - f^2$. A positive root (mode 2 on) appears only once $f^2 > c_0$, so **f_par = $\sqrt{c_0}$ = 16.77**. (If b were negative, the threshold would be $\sqrt{(c_0 - b^2/4)}$ instead.)
- Cross-check: f_par is the drive that lifts mode 1 to its saturation level, $3.288 \times \sqrt{26} = 16.77$.
- Above threshold, $x = [-b + \sqrt{(b^2 - 4(c_0 - f^2))}]/2$. At f = 17.5: $x = (-33.5 + 34.968)/2 = 0.734$, so the normalized mode-2 amplitude is 0.857 and the physical amplitude is 0.857/3.768 = 0.227.
- Just above threshold $x \approx (f^2 - c_0)/b$, so mode 2 grows like $\sqrt{(f - f_par)}$.

Accuracy near threshold. Eq. 7 relies on $\omega_1^2 - \omega_D^2 \approx -2\omega_D\gamma_1\Delta_1$. At $Q_1 = 300$ that shifts c_0 up by a fixed 3.66 (to 284.78; exact f_par ≈ 16.88). The error in x is nearly constant (about 0.10), so it matters most when x is small:

f	Eq. 7 mode-2 amplitude u_2	Envelope-equation u_2	Gap
17.0	0.128	0.094	27%
17.5	0.227	0.210	8%
18.0	0.295	0.282	4%

The correction scales like 1/Q, so the gap should shrink at $Q_1 = 1000$ (not yet run). Past the Hopf point the gap grows again, because the system circles the unstable steady state instead of sitting on it (15% at f = 19).

Step 6: Stability, Eq. 10 and the existence condition

The Solution B steady state goes unstable at f_comb = 18.26 (baseline), and the system starts pulsing: that pulsing is the comb.

The stability test. Nudge the steady state; the nudge evolves like e^{λτ}, with λ the eigenvalues of a 4×4 matrix (real and imaginary parts of ψ_1, ψ_2).

1. All Re λ < 0: nudges die out, the state is stable (Case B).
2. A complex pair λ = σ ± iΩ crosses to σ > 0: nudges grow while oscillating at Ω. This is a Hopf bifurcation, and the system settles into a stable pulsing (limit cycle, Case C).

Existence condition. Asks whether any amplitude can ever trigger the instability. If it fails, there is no comb at any drive (κ = 0 and κ = −5 both fail).

```
2\Delta_1\Delta_2 \le -\left(1 + \Delta_1^2 + 2\gamma_{21}\right)
```

Baseline: 2(5)(−6.5) = −65 ⩽ −28, so a comb is possible.

Eq. 10. The closed-form result of the stability test: the mode-2 amplitude where σ crosses zero.

```
|\psi_{2P}|^2 \ge x_{min} = -\frac{\gamma_{21}
(1+\Delta_1^2)\left[1+\Delta_1^2+4\gamma_{21}(1+\gamma_{21})\right]}
{4(1+\gamma_{21})^2\left(1+\Delta_1^2+2\gamma_{21}+2\Delta_1\Delta_2\right)}
```

- Baseline: numerator −26 × 34 = −884; denominator 16 × (26 + 2 − 65) = −592; x_min = 1.4932.
- The denominator's bracket is the existence condition: if it is positive, x_min is negative and unreachable.
- Plug x_min into Eq. 7 and solve for f: f_comb² = 1.4932² + 33.5 × 1.4932 + 281.125 = 333.377, so **f_comb = 18.26**.

What the simulation shows at the threshold.

- Onset: the full ODE starts pulsing at f ≈ 18.46; linearized envelope equations give 18.32; the paper's formula 18.26. The gap is the same finite-Q effect as in Step 5.
- Supercritical: the envelope ripple grows from zero like √(f − f_c) (7.9% at 18.5, 14.6% at 18.6, 28.2% at 19.0).
- Comb spacing at onset = Ω/2π = 0.072 Hz (period 13.9 s), matching the measured 13.84–13.95 s. Further above, the spacing changes (0.125 Hz at f = 20).

- Critical slowing down: near f_c the decay rate $-\mathrm{Re}\,\lambda \to 0$, so transients take hundreds of seconds to die out instead of about $1/\gamma_1 \approx 10$ s.

Region B width depends on drive frequency. With $\kappa = -9$ and $\gamma_{21} = 1$, the comb exists only for $0.17 \leqslant \Delta_1 \leqslant 8.83$ (f_comb $\to \infty$ at both edges, giving the U-shaped red curve of Fig. 2):

Δ_1	f_par	f_comb	Width of region B
1	6.05	6.13	0.08
3	11.96	12.30	0.34
5	16.77	18.26	1.49
7	19.76	27.01	7.25
8	20.55	43.34	22.8
9	20.87	∞	no comb

Summary: where each parameter appears

Only f, Δ_1, Δ_2 and γ_{21} move the thresholds; f_1, Q_1 and α only change units.

Parameter	Envelope eqs	Eq. 7	Eq. 10	Physical effect
f	drive term	the $-f^2$ term	—	moves you up and down Fig. 2
Δ_1	mode 1 detuning	b, c_0	yes	moves you sideways on Fig. 2
Δ_2 (via κ)	mode 2 detuning	b, c_0	yes	decides whether a comb can exist at all
γ_{21}	mode 2 damping	b, c_0	yes	lossier mode 2 raises both thresholds (5 → f_par 20.91, f_comb 25.06)
f_1, Q_1, α	gone	gone	gone	change Hz, seconds and force only